

MRIDUL KALRA

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EDUCATION

University Institute of Technology, RGPV

B.Tech, Information Technology

2022 – 2026

Bhopal, India

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, C++, SQL, HTML5, CSS3

Frontend: React.js, Next.js, React Native, Redux, Tailwind CSS

Backend: Node.js, Express.js, Django, FastAPI, WebSockets

Databases: MongoDB, PostgreSQL, MySQL, Redis

Cloud & DevOps: AWS (EC2, S3, Lambda), Docker, Jenkins

AI/ML: Google Gemini API, Pinecone Vector Database, LangChain

PROFESSIONAL EXPERIENCE

Full Stack Software Engineer Intern

Jun 2024 – Dec 2024

Aeonion Devs (Early-Stage Startup)

Remote

- Architected and developed full-stack mobile application **Rishteyy** using React Native and Node.js/MongoDB, implementing JWT authentication and real-time chat via WebSockets – scaling the app to **20,000+ downloads**
- Achieved **99.9% production uptime** and sustained **15,000+ daily API requests** by deploying Rishteyy on **AWS EC2** with Docker containerization, enabling zero-downtime rollouts and consistent cross-environment builds
- Accelerated backend API performance by engineering a **Redis caching** strategy and redesigning MongoDB collection schemas with compound indexes – reducing average response time by **73%** (450ms to 120ms) and cutting query execution time by **65%** across 8 critical endpoints
- Engineered **Campusly**, a student marketplace platform, from 0 to **2,000+ active users** by building a Node.js/React full-stack architecture with real-time product listings, user profiles, and full-text search

TECHNICAL PROJECTS

AI Drug Repurposing Platform – EY Techathon 6.0 Round 1 Winner (2024)

Stack: Python, Django, FastAPI, MongoDB, React, AWS, Redis

Live Demo

- Architected full-stack biomedical research platform enabling researchers to analyze drug repurposing opportunities through 3 integrated external data sources (PubMed, ClinicalTrials) via Django microservices backend and React frontend
- Reduced API response times by **60%** and enabled real-time analysis across **500+ compound records** by implementing a Redis caching layer for repeated medical research queries
- Secured **1st place in Round 1** of EY Techathon 6.0 among **10,000+ participants** by deploying on AWS with automated CI/CD pipeline and achieving **85%+ test coverage**

AI Knowledge Management with Vector Search

Stack: React, TypeScript, Node.js, MongoDB, Pinecone, Gemini AI

Live Demo

- Built a semantic document search engine enabling natural language queries across **1,000+ documents**, achieving sub-300ms latency by integrating **Pinecone** vector embeddings with **Google Gemini API** for context-aware retrieval
- Designed RESTful APIs with Express.js and TypeScript; optimized MongoDB performance via compound indexing and connection pooling, supporting **50+ concurrent users** with **sub-200ms** response times

Automated CI/CD Pipeline with Docker & Jenkins

Stack: Jenkins, Docker, AWS EC2, Shell Scripting

GitHub

- Reduced deployment time by **87.5%** (from 40 minutes to under 5 minutes) and eliminated environment-related release failures by engineering a Jenkins-based CI/CD pipeline with automated build, test, and Docker-containerized deployment workflows on AWS EC2
- Configured automated build triggers on Git commits with integrated testing phases, achieving **100% automated deployment** for all feature branches and enabling zero-downtime rollbacks

ACHIEVEMENTS & CERTIFICATIONS

Competitive Programming:

- LeetCode **1700+ Rating** (1000+ problems solved) | CodeChef **3-Star** (1600+ rating)

Hackathons:

- TCS CodeVita **Global Rank 636** | Flipkart GRID **Semifinalist** | EY Techathon 6.0 **Round 1 Winner** (2024)

Certifications:

- AWS Certified Cloud Practitioner
- Generative AI with LLMs – IIT Madras (NPTEL)
- Machine Learning – IIT Kharagpur (NPTEL)